

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457499.72926 +/- 0.00016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.044
Transit depth (Rp/Rs)^2	2.29 +/- 1.34 [%]
Orbital Inclination (inc)	83.79 +/- 2.86
Airmass coefficient 1 (a1)	0.49 +/- 0.28
Airmass coefficient 2 (a2)	0.38 +/- 0.28
Scatter in the residuals of the lightcurve fit is	57.53 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	3.75
Transit Duration (day)	0.07 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.044 vs 0.1214 (deviation 0.54σ)
Duration fit vs published	0.07 d vs 0.0742 d (deviation 0.12σ)
Mid-time O–C	-0.5 min
Depth SNR	2.7
Residual scatter	57.53 %

Light curve

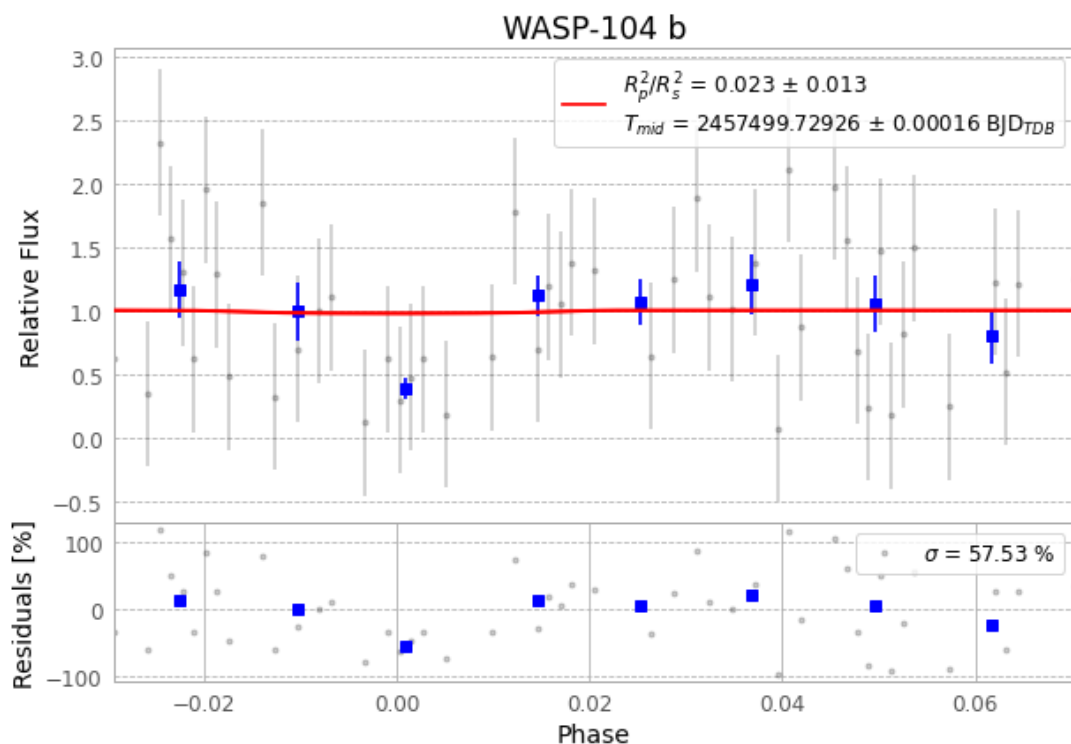


Figure 1 — FinalLightCurve_WASP-104 b_2016-04-20.png (SHA-256 b62178a653e2c05f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [510, 56], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bc6e6c3fb8020bbd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

88 FITS frames (58 MB), WASP-104160421040912.FITS → WASP-104160421083012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-160421.log (a25ea5299066...), FinalLightCurve_WASP-104 b_2016-04-20.png (b62178a653e2...), FinalLightCurve_WASP-104 b_2016-04-20.csv (b084f5b84bd4...)

Compute resources

Wall time 128.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 00:31Z.